

Contact

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(LinkedIn)

Top Skills

Jumeaux numériques
Vision par ordinateur
Intelligence artificielle (IA)

Languages

Chinois (Elementary)
Français (Native or Bilingual)
Anglais (Full Professional)
Espagnol (Elementary)

Certifications

International English Language
Testing System (IELTS)
Premier Secours en Santé Mentale

Honors-Awards

Premier prix WebCup junior Réunion

John Hoarau

Engineering Student (ESTIA/Cranfield) - R&D Intern @ALTEN -
FEA, CFD, CAD, Python, Digital Twins & AI Vision
Greater Toulouse Metropolitan Area

Summary

As an engineering student completing a dual MEng/MSc in Computational Software Techniques at ESTIA and Cranfield University, I combine physics-based simulation (FEA/CFD) with AI-driven methods to solve complex engineering challenges with a focus on aerospace, space, and defence applications. Recent highlights: - Currently an R&D Engineer at ALTEN SO (Digital Twins & AI team), building computer-vision and predictive-modelling solutions for supply chain and satellite-imagery applications. - Engineered the structural design and performed FEA for 7 industrial steel buildings valued at over \$2.2M, using ANSYS and Autodesk Robot. - Developed a vision model non-contact cardiac monitoring system in Python, to extract vital signs from video with 81% accuracy. Key technical proficiencies: - Simulation & Analysis: ANSYS, MATLAB/Simulink, CFD, FEM, ROBOT Structural Analysis - Programming & AI: Python (PyTorch, TensorFlow, Scikit-Learn, OpenCV, SciPy), C++, optimisation algorithms (PSO, Simulated Annealing, BFGS) - CAD & 3D Modelling: CATIA V5/V6, SolidWorks, OnShape I complete my internship in October 2026 and am actively looking for a role in aerospace, space, or defence engineering. Feel free to connect or message me if you're hiring for simulation, R&D, or engineering-AI roles. I'd love to hear from you.

Experience

ALTEN

R&D Engineer – Digital Twins & AI Vision
April 2026 - Present (6 months)
Toulouse

Designed and built a cognitive "adequacy score" system for a supply chain digital twin comparing an operator's declared urgency against a real urgency computed live from operational KPIs, using multi-criteria decision analysis (AHP, Fuzzy Best-Worst Method), Bayesian calibration, and risk propagation

across a multi-tier supplier network. Delivered and validated through a structured simulation exercise.

Built a computer-vision pipeline to estimate warehouse storage capacity from satellite imagery fine-tuning a YOLO segmentation model and combining it with geometric rules, to estimate storage volume, loading-dock capacity, and throughput without site visits.

Led both projects independently end-to-end, from literature review and mathematical modelling through to production-ready implementation and empirical validation.

CMR - Steel and Metal Construction Réunion Civil Engineering Intern - Structural Beam Design June 2025 - September 2025 (4 months)

La Réunion

- Engineered the structural design and performed Finite Element Analysis (FEA) for 7 industrial steel buildings valued at over \$2.2M, ensuring full compliance with Euro-Code standards.
- Ran static, dynamic, cyclic-response, and extreme-wind-load FEA simulations in ANSYS and Autodesk Robot to optimise S235/S355 steel profiles
- Authored detailed calculation reports and gained proficiency with AutoCAD and Tekla Structures for technical deliverables.

KOMPAÏ robotics Intern in Digital Project Management and CMR Development January 2023 - March 2023 (3 months)

Bidart, Nouvelle-Aquitaine, France

- As part of my test development, I created a comprehensive testing process for an autonomous robot using Python. I captured over 50 hours of operational data, which led to the identification and resolution of 2 critical software bugs.
- I managed the deployment and configuration of the Odoo platform for Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM). Through this, I automated key sales workflows and reduced manual data entry for the team by 20%.

- I executed a web performance management project where I re-engineered front-end assets and database queries. This initiative resulted in a 28% decrease in average page load times for the company's commercial websites.

Education

Cranfield University

Master of Science - MSc Computational Software Techniques

Engineering, Computational Software Techniques Engineering · (October 2025 - June 2026)

ESTIA - École Supérieure des Technologies Industrielles Avancées

Master of General Engineering, Ingénierie · (January 2022 - September 2026)

Vysoká škola báňská - Technická univerzita Ostrava

ERASMUS, Nuclear Engineering · (February 2025 - June 2025)

Lycée Amiral Bouvet

Ecole préparatoire au grande école, Calcul scientifique · (August 2020 - July 2022)

Lycée polyvalent de Trois-Bassin

Baccalauréat sciences et technologies de l'industrie et du développement durable, Calcul scientifique · (August 2017 - July 2020)